6.P3 ~~SP - 5 PU VC 3~~ VC - 63 SP - 90 FC 135000

Current monthly sales / SU = 8000 lanterns

1. BEP in unit ⇒ Sales = VC + FC + profit

$\Rightarrow 90x = 63x + 135000 + 0$

$\Rightarrow x = 5000$

BEP in \$ = BEP in unit × SP = 5000 × 90 = 450,000

Q3. CM Income statement

	PU	Total
Sales	90	720,000
(-) VC	63	504,000
CM	27	216,000
FC		(135,000)
NOI		81,000

SP reduce 10% | SU increase 25%.

90 - 10% = 81 8000 + 25% = 10000

Q. CM Income statement

Sales	81	810,000
(-) VC	63	630,000
CM	18	180,000
FC		(135,000)
NOI		45,000

4. Req unit = $\frac{\text{FC + Target net income}}{\text{CM per unit}}$

$= \frac{135000 + 72,000}{18} = 11,500$

(6-4) ① VC = ~~\$60~~ \$42 fc = \$450000 SU = 30,000

SP = 60 (PU)

CM Income statement

	Incom. PU	Statement total
Sales	\$ ~~30~~, 60	\$ 1800000
(-) VC	\$ 42	(\$126.0000)
CM	\$ 18.	\$ 54,0000
(-) FC		(450000)
NOI		\$ 90000

(CM = SP - VC = 18)

DOL/OL = CM/NOI = $\frac{\cancel{90000}\ 18 \times 30000}{90000}$

= 6

② (a) Increasing NOI in % = In. sales × OL

= 25% × 6

= 150%.

(b) NOI increase = 90,000 × (1+1.5)

= 225,000

(Ans)

(6.5) $SP_A = 700,000$ $SP_B = 300,000$

VC_A: $700000(1-60\%) = \$280,000$

VC_B: $300,000\cdot(1-70\%) = \$90,000$

$FC = 598,500$

Q(1)

CM Income Statement

	Income PU	~~Statement total~~
Sales	(700000+300,000)	~~1000000~~ 1000000
VC	(280,000 + 90,000)	630,000
CM		630,000
- FC		~~370,000~~ 598500
		~~630,0000~~
		31,500

(2) ~~CM = CM ratio~~ CM ratio = $\frac{CM}{Sales}$

$= \frac{31500}{1000000} = 63\%$

break event point = $\frac{FC}{CM\ ratio} = 950,000$

Writing should be started right from here

(3) · Dncreasing NODint = 50,000 x cm ratio

= 31500

6.6 1. BEP in unit

Sales = VC + FC + Profit

$\Rightarrow 40x = 28x + 150{,}000$

$\Rightarrow x = \frac{150{,}000}{12} = 12{,}500$ unit

BEP in \$ = BEP in unit × SP

= 12,5000 × 40

= 500,000

SP = 40
VC = 28
FC = 12
FC = 150,000
VC = 420,000
CM = 40 − 28
= 12

2.

3. Profit = 18,000

$\text{BEP in Unit} = \frac{\text{FC} + \text{Profit}}{\text{CM per unit}}$

$= \frac{150{,}000 + 18000}{12}$

= 14000 unit

CM Income statement

	PU	Total
Sales	40	5,60,000
(−) VC	28	392,000
CM	12	168,000
(−) FC		150000
NOI		18,000

Writing should be started right from here

4. MOS = Actual sales (SU) − BEP

$= 600{,}000 - 500{,}000$

$= 100{,}000$

MOS in % = $\dfrac{\text{MOS in \$}}{\text{A. Sales / SU}} = \dfrac{100{,}000}{600{,}000}$

$= 16.67\%$

5. CM ratio = $\dfrac{CM}{Sales} = \dfrac{180{,}000}{600000}$ → (SU given)

$= 30\%$

(6.7) cm-ratio = 40% SP = $60 FC = 360,000

(1) vc = ?

$$\text{cm ratio} = \frac{\text{cm}}{\text{sales}}$$

cm = cm ratio × sales

= 40% × 60 = 24 $

~~Sales = vc +~~ cm = SP − vc

vc = $60 − $24 = $36 (Ans)

(H) (2) NO :

(a) BEP in Unit :

(In $) sales = vc + FC + Profit

⇒ 60x = 36x + 360,000

⇒ x = 15,000 unit

(In £) BEP in $ = BEP in unit × SP

= 15,000 × 60

= 9,00,000

(b) Profit = 90,000 (Sales level ---)

(B) sales in Unit ~~In $~~ :

sales = vc + FC + profit

60x = 36x + 360,000 + 90,000

x = 18,750 unit

Dn($): sales $ = sales in unit × SP
= 18750 × 60 = 1,125,000

(b) ~~New cm per unit = $60 − $3 = $33~~
~~New cm~~

New CM = $60 − ($36 − $3) = $27

Sales = VC + FC + profit
⇒ $60x = 33x + 360,000
x = 13333.33 unit

(3) ~~OWN~~ New BEP in $ = (BEP in unit) × 60
= 799999.8 Ans

(6.9)

1.a) CM ratio = $\frac{CM}{Sales} = \frac{81,000}{270,000}$
= 30%

BEP in $ = $\frac{FC + Profit}{CM\ ratio}$
= $\frac{90,000 + 0}{30\%}$
= 300,000

3. Repeat(2) (CM...)

(a) BEP units = $\frac{\cancel{150}\cdot 360000}{24} = 15{,}000$

· ~~sales = vet Pet Profit~~

⇒ ∴ ~~sales~~ · BEP in \$ = 15,000 × 60 = 900,000

(b) · Profit = 90000

BEP in unit = $\frac{\text{FC} + \text{Profit}}{\text{CM in unit}} = \frac{360000 + 90000}{24}$

$\Rightarrow 18750$

∴ \$ = BEP in unit × SP = 18750 × 60

$\Rightarrow 1125{,}000$

(c) New Break-even (CM = 27)

BEP in units $= \frac{360{,}000}{27} = 13{,}333.3$

BEP in \$ = 13333.3 × 60 = \$ 799,999.8